

Fully Test-time Adaptation for Tabular Data

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Abstract

Tabular data plays a vital role in various real-world scenarios and finds extensive applications. Although recent deep tabular models have shown remarkable success, they still struggle to handle data distribution shifts, leading to performance degradation when testing distributions change. To remedy this, a robust tabular model must adapt to generalize to unknown distributions during testing. In this paper, we investigate the problem of fully test-time adaptation (FTTA) for tabular data, where the model is **adapted** using only the testing data. We identify three key challenges: the existence of label and covariate distribution shifts, the lack of effective data augmentation, and the sensitivity of adaptation, which render existing FTFA methods ineffective for tabular data. To this end, we propose the *Fully Test-time Adaptation for Tabular data*, namely FTAT, which enables FTFA methods to robustly optimize the label distribution of predictions, adapt to shifted covariate distributions, and suit a variety of tasks and models effectively. We conduct comprehensive experiments on six benchmark datasets, which are evaluated using three metrics. The experimental results demonstrate that FTAT outperforms state-of-the-art methods by a margin.

1 Introduction

Tabular data (Altman and Krzywinski 2017) plays a vital role in numerous practical applications, including economics (Salehpour and Samadzamini 2024), healthcare (Ching et al. 2018), finance (Ozbayoglu, Gudelek, and Sezer 2020), and manufacturing (Hein et al. 2017). Deep neural networks (DNNs) have recently shown remarkable success in handling tabular data, often surpassing traditional statistical methods when training and test data share the same distribution (Arik and Pfister 2021; Gorishniy et al. 2021). However, real-world applications often experience shifts in data distributions during testing, leading to significant performance degradation in existing methods (Kolesnikov 2023).

To address distribution shifts during testing, **fully test-time adaptation (FTFA)** algorithms have emerged, **enhancing the performance of pre-trained DNNs using only testing**

data. These methods are particularly designed to deal with **covariate distribution shift** (Wang et al. 2021), **label distribution shift** (Wu et al. 2021), or both (Zhou et al. 2023), adapting the model parameters (Wang et al. 2022) or optimizing the predictions (Boudiaf et al. 2022). **However, they are primarily designed for image tasks and heavily rely on image augmentation strategies** (Wang et al. 2022) and image-specific data assumptions (Boudiaf et al. 2022; Zhou et al. 2023), rendering them less effective for tabular data. As a result, fully test-time adaptation for tabular data remains underexplored, despite its significance in real applications.

To this end, we study the fully test-time adaptation problem setting for tabular data, namely *AdaTab*, which holds significant practical value (Altman and Krzywinski 2017). For example, in financial applications (Kritzman, Page, and Turkington 2012), the non-stationary financial market environment can cause significant changes in the data distribution between training and testing. For instance, shifts in the stock market can significantly affect market behavior and investor sentiment. These distribution shifts degrade the model performance and seriously affect investment decision-making and risk management, thereby leading to financial losses (Guo, Hu, and Yang 2023). The goal of the *AdaTab* problem setting is to adapt the trained deep tabular model to unknown distributions using only testing data, preventing the performance degradation caused by distribution shifts in downstream tabular applications.

In this paper, we conduct an in-depth investigation into the *AdaTab* problem. Our four observations reveal **three key challenges** in designing FTFA methods for tabular data: (a) Covariate and label distribution shifts exist in tabular data, but they cannot be effectively addressed by existing FTFA methods; (b) Typical augmentation for test-time adaptation is often ineffective for tabular data, limiting the ability of FTFA methods to compute consistency; (c) Adaptation is sensitive to both tasks and models for tabular data. To address these challenges, we propose a novel FTFA approach, FTAT. It comprises **three essential modules**: *Confident Distribution Optimizer*, *Local Consistent Weighter*, and *Dynamic Model Ensembler*, which robustly track and optimize the label distribution of predictions, adapt the model to shifted covariate distribution, and dynamically adapt the model for various tasks and models. To summarize, the contributions of this paper are threefold:

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- (1) We investigate fully test-time adaptation for tabular data, identifying **three key challenges: the existence of label and covariate distribution shifts, the lack of effective data augmentation, and the sensitivity of adaptation.**
- (2) We propose a novel approach, FTAT, which incorporates the **Confident Distribution Optimizer, Local Consistent Weighter, and Dynamic Model Ensemble** to address the challenges of shifted label distribution, shifted covariate distribution, and sensitivity, respectively.
- (3) We **evaluate** the FTAT approach on six tabular benchmarks with real distribution shifts using three backbone models, demonstrating that the proposed approach significantly outperforms state-of-the-art FTTA methods.

2 Problem and Analysis

In this section, we first introduce the *AdaTab* problem setting, including the notations and problem formulation. We then present four observations of *AdaTab*, which highlight three main challenges and underscore the necessity of designing FTTA methods specifically for tabular data.

2.1 Problem Formulation

We consider the fully test-time adaptation problem setting for tabular classification problem, namely, *AdaTab*. The input space is $\mathcal{X} \in \mathbb{R}^d$, where d is the number of features. Each feature can be a continuous or discrete value. The label space is $\mathcal{Y} \in \{0, 1\}^K$, where the K is the number of classes. In this setting, we are given a well-trained source tabular model $f_{\theta_0} : \mathcal{X} \mapsto \mathcal{Y}$ with the initial parameters θ_0 . During the testing phase, the model is solely adapted based on the unlabeled batched testing data D_t at each timestamp t , updating its parameters from θ_t to θ_{t+1} . The goal of *AdaTab* problem is to adapt the initial given model f_{θ_0} during the testing phase, so that the adapted model f_{θ_t} can generalize better on the test data D_t at each timestamp t .

2.2 Problem Analysis

In the context of the *AdaTab* problem, we have identified four observations that also serve as key challenges hindering FTTA methods from effectively working with tabular data, in contrast to the standard fully test-time adaptation designed for image tasks.

Observation 1: Covariate distribution and label distribution shifts in tabular data hinder performance of FTTA methods. Our first observation reveals that both covariate distribution and label distribution shifts exist in tabular data, and both contribute to performance degradation. To estimate the distribution shifts between the training and testing datasets, we use the optimal transport dataset distance with Gaussian approximation (Alvarez-Melis and Fusi 2020) to measure covariate distribution shifts and the L_2 distance (Gardner, Popovic, and Schmidt 2023) to assess label distribution shifts. As shown in Fig. 1, both increases in label distribution shift (DIABETE→HELOC) and covariate distribution shift (HELOC→ASSIST) degrade performance on the testing data with distribution shifts. However, as our experimental results reveal, existing robust FTTA methods,

Table 1: Performance of the non-adaptation baseline and CoTTA method with different augmentation strengths σ using the MLP model. The best performance is in **bold**.

Method	DIABETE	HELOC	ASSIST
Non-Adaptation	60.82 ± 0.22	54.37 ± 5.35	55.86 ± 3.81
$\sigma = 0.2$	60.46 ± 0.20	46.40 ± 3.08	54.89 ± 1.88
$\sigma = 0.4$	59.18 ± 0.42	43.36 ± 0.25	54.86 ± 3.00
$\sigma = 0.6$	57.73 ± 0.64	43.06 ± 0.07	54.51 ± 2.26
$\sigma = 0.8$	56.19 ± 0.83	43.07 ± 0.03	53.79 ± 3.80
$\sigma = 1.0$	54.74 ± 0.77	43.09 ± 0.01	54.23 ± 3.56

such as ODS (Zhou et al. 2023), designed to address covariate and label distribution shifts in tabular data, do not perform well in practice. This observation highlights the challenges faced by FTTA methods designed for tabular data in addressing both covariate and label distribution shifts simultaneously. We also estimate the label distribution of data whose model prediction entropy is lower than thresholds on DIABETES dataset. The results in Fig. 2 demonstrate that we can still accurately estimate the label distribution using only low-entropy data.

Observation 2: Typical Augmentation used in test-time adaptation is ineffective for tabular data. Existing FTTA methods, such as CoTTA (Wang et al. 2022) and AdaContrast (Chen et al. 2022), rely heavily on data augmentation. However, augmentation for tabular data is not as effective as it is for images. We conduct experiments based on CoTTA methods with perturb augmentation (Fang et al. 2022) with different perturbation strengths controlled by σ . As shown in Table 1, the performance of CoTTA degrades as the augmentation strength increases, and it fails to surpasses the non-adaptation baseline. This observation highlights the challenges of FTTA methods designed for tabular data, particularly their inability to rely on data augmentation method when dealing with tabular data.

Observation 3: Adaptation is sensitive to both tasks and models for tabular data. Unlike images, which exhibit strong transferability (He, Girshick, and Dollár 2019) and similar structure (Torralba and Oliva 2003), tabular data from different tasks differs significantly. Moreover, different backbone models (Gorishniy et al. 2021; Huang et al. 2020) are designed to address distinct tasks. Therefore, for specific tabular tasks and the backbone models used, FTTA methods require tuning for optimal performance. As shown in Fig. 3, the optimal learning rates for different backbone models on the same task and same backbone model on different tasks varies. This observation indicates that the *AdaTab* problem requires a model capable of dynamically tuning the learning rates, rather than relying on a fixed learning rate.

Observation 4: Existing FTTA methods degrades when dealing with tabular data. Observations 1, 2, and 3 also serve as key challenges in designing FTTA methods for tabular data, causing existing FTTA methods to fail to improve performance compared to the baseline. As shown in Tab. 2, we compare performance of non-adaptation baseline with

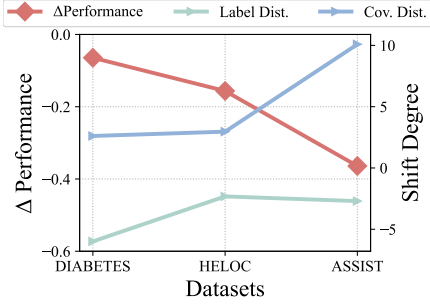


Figure 1: The label and covariate distribution shifts between training and testing in tabular data degrade the model performance. The shift degree is taken logarithm for aesthetic purposes.

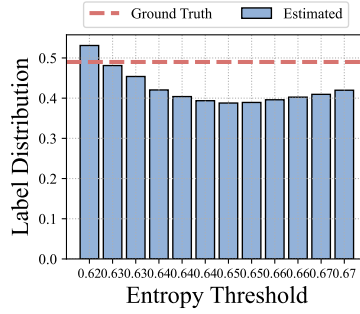


Figure 2: The label distribution of data for which the entropy of model predictions is lower than various thresholds. The ground truth is marked with dashed line.

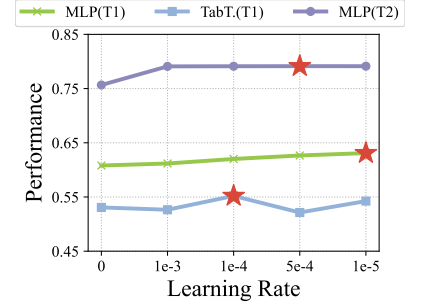


Figure 3: The performance of FTAT with different learning rates. The optimal value differs across backbones and tasks. The highest point of each line is marked by a red star.

Table 2: Performance of the non-adaptation baseline and two representative FTTA methods using an MLP backbone model. Degraded performance is underlined.

Method	DIABETE	HELOC	ASSIST
Non-Adaptation	60.82 ± 0.22	54.37 ± 5.35	55.86 ± 3.81
Optimize Parameters	61.34 ± 0.33	<u>54.35 ± 5.38</u>	<u>50.87 ± 0.32</u>
Optimize Predictions	61.47 ± 0.35	<u>43.10 ± 0.00</u>	<u>45.12 ± 0.18</u>

two representative FTTA methods which respectively optimize model parameters and predictions, i.e., TENT (Wang et al. 2021) and LAME (Boudiaf et al. 2022). As the covariate distribution and label distribution shifts become more severe (DIABETE→HELOC→ASSIST), both types of FTTA methods fail to surpass the non-adaptation baseline. This observation highlights the significance of developing FTTA methods specifically designed for tabular data which addressing three challenges simultaneously.

3 Methodology

In this section, we introduce our FTAT approach for *AdaTab* problem setting. As discussed in the analysis section, the *AdaTab* problem encompasses three challenges:

- Covariate and label distribution shifts exist in tabular data, but cannot be effectively addressed by existing FTTA methods;
- Typical augmentation used for test-time adaptation is not very effective for tabular data, limiting the ability of FTTA methods to adopt consistency;
- Adaptation is sensitive to both tasks and models for tabular data.

To address the above challenges, we introduce three modules specifically designed for *AdaTab* problem, i.e., *Confident Distribution Optimizer*, *Local Consistent Weighter*, and *Dynamic Model Ensembler*. Specifically, the *Confident Distribution Optimizer* optimizes the original model predictions $f_{\theta_t}(x)$ to $\hat{f}_{\theta_t}(x)$ for a data point x at timestamp t . The *Local*

Consistent Weighter affects the adaptation objective:

$$\theta_{t+1} = \arg \min_{\theta} \sum_{i=1}^{D_t} \mathcal{W}(x_i, D_t, \theta_t) \cdot \text{Loss}(\hat{f}_{\theta_t}(x_i)) \quad (1)$$

using a weighting function $\mathcal{W}$, where $\text{Loss}(\cdot)$ represents the unsupervised loss for test-time adaptation, and we employ entropy loss in accordance with classical methods. The *Dynamic Model Ensembler* maintains multiple models and ensembles their predictions in an online manner. We will introduce them in detail.

3.1 Confident Distribution Optimizer

First, we aim to optimize the model predictions to align with the current shifted label distribution. The existing solution (Zhou et al. 2023) fails because the challenging nature of tabular data prevents the model from making accurate predictions, which in turn hinders the estimation of the label distribution. Therefore, the key challenge is how to robustly track the shifted label distribution $\hat{P}_t$ at each timestamp t . With original label distribution P_0 and estimated label distribution $\hat{P}_t$, the optimized model prediction $\hat{f}_{\theta_{t+1}}(x_k)$ for next timestamp on data point x_k is

$$\hat{f}_{\theta_{t+1}}(x_k) = \frac{f_{\theta_{t+1}}(x_k) \circ \hat{P}_t}{P_0} \quad (2)$$

Motivated by our observations, we recognize that we can estimate the label distribution $\tilde{P}_t$ with bias from model $\hat{f}_{\theta_t}$ at each timestamp t using only data with low-entropy predictions (i.e., data with confident predictions):

$$\tilde{P}_t = \frac{\sum_{i=1}^{|D_t|} \mathbb{I}[\text{Entropy}(\hat{f}_{\theta_t}(x_i)) < \epsilon] \cdot \hat{f}_{\theta_t}(x_i)}{\sum_{i=1}^{|D_t|} \mathbb{I}[\text{Entropy}(\hat{f}_{\theta_t}(x_i)) < \epsilon]} \quad (3)$$

where D_t is current data batch, ϵ is a threshold, and $\text{Entropy}(\cdot)$ is the function for computing entropy of predictions. Note that there exists bias in estimated $\tilde{P}_t$ as the model predictions may contains errors. To address this issue, we

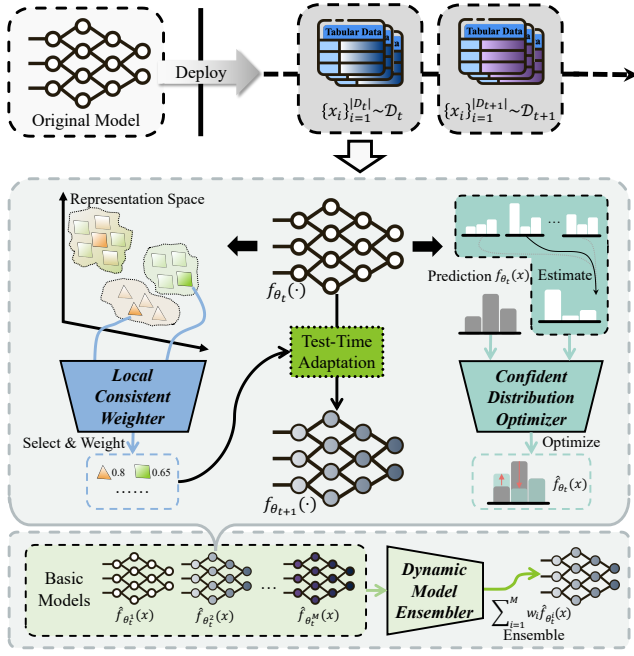


Figure 4: The overall illustration of FTAT approach.

compute the covariate matrix $\hat{C}_t$ at the current timestamp t , where its k -th row is equal to

$$\frac{\sum_{i=1}^{|D_t|} \mathbb{I} \left[\arg \max_j \hat{f}_{\theta_t}(x_i)_j = k \right] \cdot \hat{f}_{\theta_t}(x_i)}{\sum_{i=1}^{|D_t|} \mathbb{I} \left[\arg \max_j \hat{f}_{\theta_t}(x_i)_j = k \right]} \quad (4)$$

Then, the unbiased label distribution is $\hat{C}_t^{-1} \tilde{P}_t$. We additionally adopt a temperal ensemble method (Laine and Aila 2017) to robustly track estimated label distribution smoothly with a factor α and previous estimated shift $\hat{P}_{t-1}$:

$$\hat{P}_t = \text{Norm} \left(\hat{P}_{t-1} - \alpha \cdot \hat{C}_t^{-1} \tilde{P}_t \right) \quad (5)$$

where $\text{Norm}(\cdot)$ normalizes the distribution to sum to one, and we use the Softmax function for this purpose.

3.2 Local Consistent Weighter

Second, to mitigate the adverse effects of shifted covariate distribution, we propose filtering testing data with low-quality predictions to ensure robust test-time adaptation and avoiding error accumulation. However, for tabular data, computing consistency through data augmentation is non-trivial because our observation indicates that augmentation for tabular data is not as reliable as it is for image data.

To address this issue, we propose replacing the consistency between a data point and its augmentations with the consistency between a data point and its neighborhood, under the inspiration of one existing tablar study (Gorishniy et al. 2024). Specifically, we define the neighborhood set $N(x_k, D_t)$ of each data point x_k in current batch D_t measured by one distance function $\text{Dist}(\cdot, \cdot)$:

$$N(x_k, D_t) = \{x | \text{Dist}(x, x_k) < \overline{\text{Dist}}_t, x \in D_t\} \quad (6)$$

where $\overline{\text{Dist}}_t = \frac{2}{|D_t|(|D_t|-1)} \sum_{i=1}^{|D_t|} \sum_{j=i+1}^{|D_t|} \text{Dist}(x_i, x_j)$ is the average pair-wise distance in D_t and we adopt L_2 distance as the distance function. Next, we define the prediction of one data point x_k is consistent if its soft pseudo-label vector is close to the average soft pseudo-label vectors in its neighborhood $N(x_k, D_t)$. Then, we define the indication function $\mathcal{I}(x_k, D_t, \theta_t)$ to decide whether one data point x_k in current batch D_t is consistent:

$$\mathcal{I}(x_k, D_t, \theta_t) = \begin{cases} 1, & \left\| f_{\theta_t}(x_k) - \frac{\sum_{x \in N(x_k, D_t)} f_{\theta_t}(x)}{|N(x_k, D_t)|} \right\| < \beta, \\ 0, & \text{Otherwise.} \end{cases} \quad (7)$$

where θ_t is the parameters of model at timestamp t , $f_{\theta_t}(\cdot)$ predicts pseudo-label of data point, and β is a hyperparameter to control the degree of consistency. To additionally ensure the robustness of adaptation, we compute the uncertainty of each data point using margin of prediction (Helton and Johnson 2011) as $\max \hat{f}_{\theta_t}(x_k) - \min \hat{f}_{\theta_t}(x_k)$. Finally, our proposed local consistent weighter $\mathcal{W}(x_k, D_t, \theta_t)$ is formulated as follows:

$$\left[\max \hat{f}_{\theta_t}(x_k) - \min \hat{f}_{\theta_t}(x_k) \right] \cdot \mathcal{I}(x_k, D_t, \theta_t) \quad (8)$$

3.3 Dynamic Model Ensemble

Third, to address the sensitivity issue of adaptation, we employ the online ensemble learning paradigm (Bai et al. 2022) to optimize multiple models with different learning rates and ensemble their outputs through weighted averaging to obtain the overall robust prediction. Specifically, we maintain M models using different learning rates during the testing, denoted as the set of base models $\{\hat{f}_{\theta_i}\}_{i=1}^M$. Then, model predictions are weighted according to corresponding loss values $w_i \propto 1 - R_t^i(D_t)$, where $R_t^i(D_t)$ is the loss value of i -th model $\hat{f}_{\theta_i}$ evaluated on current batched data D_t and satisfies the constraint $\sum_{i=1}^M w_i = 1$. The final prediction of the FTAT approach is obtained by weighted ensemble for a data point x , that is, $\sum_{i=1}^M w_i \cdot \hat{f}_{\theta_i}(x)$.

4 Experiments

In this section, we first introduce the experimental setup. Next, we present our empirical results, comparing our FTAT approach with existing FTFA methods across four benchmark datasets. Finally, we conduct an ablation study and provide further analysis for our proposed method.

4.1 Experimental Setup

Evaluation Protocol. In our experiments on tabular tasks, we follow the fully test-time adaptation setting, where the source model is trained on training data and adapted to shifted test data without any access to the source training data. Specifically, we train the source model on training data and select the best model based on the validation set following the TableShift benchmark (Gardner, Popovic, and Schmidt 2023). Then, FTAT approach and existing FTFA methods are evaluated on the shifted test set. We select six common tabular benchmark datasets from the TableShift

Table 3: The average performance of FTAT approach and comparison methods using three backbone models. The best performance is in **bold**. Our FTAT approach achieves the best performance across all three backbone models.

Method	MLP			TabTransformer			FT-Transformer		
	Acc.	Balanced Acc.	F1	Acc.	Balanced Acc.	F1	Acc.	Balanced Acc.	F1
Non-Adaptation	62.45	64.61	60.59	60.86	63.08	58.32	59.69	62.49	54.29
TENT	58.43	61.63	50.97	58.32	61.40	51.73	51.61	55.41	36.34
EATA	61.43	63.69	60.11	60.33	62.36	60.09	56.04	58.95	44.62
LAME	59.48	62.32	58.47	59.15	62.50	58.39	58.90	61.98	51.86
CoTTA	61.59	63.78	60.57	60.37	62.82	59.75	59.64	62.43	53.41
ODS	59.18	62.15	57.83	59.22	62.02	58.46	59.05	61.70	51.41
SAR	61.16	63.49	59.18	60.30	62.77	59.72	59.27	62.11	57.04
FTAT	66.77	64.96	72.00	66.14	64.40	69.03	64.01	62.54	69.56

benchmark, which exhibit significant performance gaps under distribution shifts. Therefore, these datasets contains samples from 10K to 5M and features from 26 to 365, which can cover a wide range of tabular scenarios under distribution shifts. All experiments are repeated with different random seeds, and the mean and standard deviation are reported.

Comparison Methods. To compare our FTAT approach with various FTTA methods, including typical FTTA methods (i.e., TENT (Wang et al. 2021) and EATA (Niu et al. 2022)), continual FTTA methods (i.e., CoTTA (Wang et al. 2022)), and recently proposed robust FTTA methods (i.e., SAR (Niu et al. 2023), LAME (Boudiaf et al. 2022), and ODS (Zhou et al. 2023)).

4.2 Main Results

To evaluate the effectiveness of FTAT, we report the detailed experimental results using three backbone models in Tab. 3. The performance are measured by three metrics including accuracy, balanced accuracy, and F1 score. The experimental results show our FTAT approach outperforms existing methods by a margin on all metrics.

Moreover, we report the detailed results on each dataset using MLP backbone model in Tab. 4. FTAT achieves the best performance on major cases and give competitive performance on the resting cases, demonstrating the effectiveness of FTAT on various tabular datasets with different backbone models. ODS (Zhou et al. 2023) addresses both the covariate and label distribution shifts, however, it underperforms FTAT on most cases, demonstrating the effectiveness of FTAT in handling distribution shifts for tabular data. The detailed results using FT-Transformer (Gorishniy et al. 2021) and TabTransformer (Huang et al. 2020) are included in our supplementary material due to the space limits.

Our experimental results confirm our last observation, showing that existing FTTA methods face performance degradation on tabular data with distribution shifts, thereby demonstrating the necessity to study the *AdaTab* problem. Our FTAT approach consistently outperforms non-adaptation baseline and existing FTTA methods in most cases, offering insights into this challenging problem.

4.3 Further Analysis

Ablation Study. We analyze the effectiveness of the *Confident Distribution Optimizer* (denoted as CDO) and the *Local Consistent Weighter* (denoted as LCW) in Tab. 5 on DIABETE and HELOC datasets using the MLP backbone model. The experimental results of FTAT approach without CDO and LCW are reported measured by three metrics. Without CDO, the performance of FTAT approach improves marginally, which indicates that the label distribution shift hinders the performance and CDO addressing the label distribution shift plays a more important role in the FTAT approach. Without LCW, the performance of FTAT cannot achieve the optimal level, demonstrating the essential role of LCW to robustly update the model. Overall, our FTAT approach achieves the best performance when both CDO and LCW are employed, demonstrating their effectiveness in addressing challenges of FTTA for tabular data.

Estimation of Label Distribution. We compare the performance in estimating label distribution on the DIABETE dataset using the MLP backbone model. We adopt KL divergence to measure the distance between the ground-truth label distribution and its estimation. As shown in Fig. 5, the LAME method cannot accurately estimate the label distribution. While the ODS method can robustly track the label distribution, it requires several iterations to converge to an accurate estimation. In contrast, our FTAT approach achieves accurate label distribution estimation at a much faster speed. This result demonstrates the superiority of FTAT approach.

Effects of Dynamic Model Ensembler. To validate the effectiveness of *Dynamic Model Ensembler*, we conduct experiments running with base models with different learning rates, ensemble baseline, and FTAT approach. Here, we compare with four base models with different learning rates $\{1e-3, 1e-4, 5e-4, 1e-5\}$. The ensemble baseline are the direct ensemble of these base models. 6 presents the average performance using MLP model on DIABETE dataset. The results show that our *Dynamic Model Ensembler* module consistently outperforms the average ensemble baseline, demonstrating the its effectiveness. Moreover, the FTAT approach can achieve the best performance or competitive performance when compared to base learners with the optimal

Table 4: Performance of FTAT approach and comparison methods on 6 datasets using MLP. The best performance is in **bold**.

Method	HELOC			ANES			Health Ins.		
	Acc.	Balanced Acc.	F1	Acc.	Balanced Acc.	F1	Acc.	Balanced Acc.	F1
Non-Adaptation	54.37 $\pm$ 5.35	58.25 $\pm$ 3.56	40.02 $\pm$ 16.8	79.11 $\pm$ 0.31	75.66 $\pm$ 0.46	84.24 $\pm$ 0.16	65.79 $\pm$ 0.63	70.68 $\pm$ 0.44	66.21 $\pm$ 0.90
TENT	54.35 $\pm$ 5.38	58.24 $\pm$ 3.58	39.95 $\pm$ 16.9	78.07 $\pm$ 0.35	74.09 $\pm$ 0.65	83.76 $\pm$ 0.13	64.30 $\pm$ 0.70	69.79 $\pm$ 0.47	63.87 $\pm$ 1.06
EATA	54.37 $\pm$ 5.35	58.25 $\pm$ 3.56	40.02 $\pm$ 16.8	78.13 $\pm$ 0.30	74.20 $\pm$ 0.59	83.79 $\pm$ 0.10	65.78 $\pm$ 0.63	70.68 $\pm$ 0.44	66.21 $\pm$ 0.90
LAME	43.10 $\pm$ 0.00	50.00 $\pm$ 0.00	30.10 $\pm$ 0.00	63.50 $\pm$ 0.00	54.60 $\pm$ 0.00	46.80 $\pm$ 0.00	63.44 $\pm$ 1.69	69.14 $\pm$ 1.09	62.61 $\pm$ 2.69
CoTTA	54.36 $\pm$ 5.35	58.25 $\pm$ 3.56	40.03 $\pm$ 16.8	78.13 $\pm$ 0.30	74.20 $\pm$ 0.59	83.79 $\pm$ 0.10	65.79 $\pm$ 0.63	70.68 $\pm$ 0.44	66.21 $\pm$ 0.90
ODS	43.10 $\pm$ 0.00	50.00 $\pm$ 0.00	30.10 $\pm$ 0.00	63.50 $\pm$ 0.00	54.60 $\pm$ 0.00	46.80 $\pm$ 0.00	63.45 $\pm$ 1.68	69.14 $\pm$ 1.07	62.62 $\pm$ 2.68
SAR	52.32 $\pm$ 6.05	56.74 $\pm$ 3.99	33.16 $\pm$ 19.0	78.13 $\pm$ 0.30	74.20 $\pm$ 0.59	83.79 $\pm$ 0.10	65.79 $\pm$ 0.63	70.68 $\pm$ 0.44	66.21 $\pm$ 0.90
FTAT	64.09 $\pm$ 1.14	63.64 $\pm$ 0.93	67.80 $\pm$ 2.71	80.09 $\pm$ 0.23	79.12 $\pm$ 0.20	83.42 $\pm$ 0.25	72.42 $\pm$ 0.20	65.30 $\pm$ 0.15	80.83 $\pm$ 0.23

Method	ASSIST			DIABETE			Hypertension		
	Acc.	Balanced Acc.	F1	Acc.	Balanced Acc.	F1	Acc.	Balanced Acc.	F1
Non-Adaptation	55.86 $\pm$ 3.81	60.81 $\pm$ 3.37	66.42 $\pm$ 1.86	60.81 $\pm$ 0.21	60.59 $\pm$ 0.24	51.18 $\pm$ 1.69	58.76 $\pm$ 1.68	61.69 $\pm$ 0.95	55.46 $\pm$ 4.03
TENT	50.87 $\pm$ 0.32	56.41 $\pm$ 0.29	63.99 $\pm$ 0.15	61.34 $\pm$ 0.33	61.15 $\pm$ 0.34	53.75 $\pm$ 1.01	41.67 $\pm$ 0.08	50.07 $\pm$ 0.05	0.49 $\pm$ 0.36
EATA	55.86 $\pm$ 0.18	60.81 $\pm$ 0.16	66.42 $\pm$ 0.08	61.36 $\pm$ 0.30	61.16 $\pm$ 0.31	53.68 $\pm$ 1.09	57.81 $\pm$ 2.32	61.19 $\pm$ 1.38	52.87 $\pm$ 5.82
LAME	45.12 $\pm$ 0.18	51.30 $\pm$ 0.18	61.40 $\pm$ 0.18	61.47 $\pm$ 0.35	61.30 $\pm$ 0.37	54.67 $\pm$ 1.45	58.63 $\pm$ 1.60	61.64 $\pm$ 0.92	55.12 $\pm$ 3.84
CoTTA	55.86 $\pm$ 0.18	60.81 $\pm$ 0.16	66.42 $\pm$ 0.08	61.39 $\pm$ 0.29	61.20 $\pm$ 0.30	53.82 $\pm$ 1.05	58.76 $\pm$ 1.68	61.69 $\pm$ 0.95	55.46 $\pm$ 4.03
ODS	45.12 $\pm$ 0.18	51.30 $\pm$ 0.18	61.40 $\pm$ 0.18	61.47 $\pm$ 0.35	61.30 $\pm$ 0.37	54.69 $\pm$ 1.43	57.12 $\pm$ 1.46	60.80 $\pm$ 0.93	51.41 $\pm$ 3.43
SAR	55.86 $\pm$ 0.18	60.81 $\pm$ 0.16	66.42 $\pm$ 0.08	61.38 $\pm$ 0.30	61.19 $\pm$ 0.30	53.98 $\pm$ 0.93	58.21 $\pm$ 1.51	61.50 $\pm$ 0.77	53.81 $\pm$ 4.05
FTAT	60.17 $\pm$ 2.87	63.79 $\pm$ 2.18	66.92 $\pm$ 1.20	61.66 $\pm$ 0.30	61.54 $\pm$ 0.28	59.27 $\pm$ 0.96	62.20 $\pm$ 0.94	56.36 $\pm$ 1.62	73.77 $\pm$ 0.13

Table 5: Ablation study of FTAT approach on DIABETE dataset using MLP backbone model. The best performance is in **bold**. The results show that both *Confident Distribution Optimizer* (denoted as CDO) and the *Local Consistent Weighter* (denoted as LCW) are essential for FTAT approach.

DIABETE			
Method	Acc.	Balanced Acc.	F1
Non-Adaptation	60.81 $\pm$ 0.21	60.59 $\pm$ 0.23	51.18 $\pm$ 1.69
FTAT w/o CDO	60.85 $\pm$ 0.22	60.61 $\pm$ 0.24	51.26 $\pm$ 1.69
FTAT w/o LCW	61.43 $\pm$ 0.16	61.28 $\pm$ 0.20	55.61 $\pm$ 1.67
FTAT	61.66 $\pm$ 0.30	61.54 $\pm$ 0.28	59.27 $\pm$ 0.96

HELOC			
Method	Acc.	Balanced Acc.	F1
Non-Adaptation	54.37 $\pm$ 5.35	58.25 $\pm$ 3.56	40.02 $\pm$ 16.8
FTAT w/o CDO	62.73 $\pm$ 0.23	62.55 $\pm$ 0.52	66.45 $\pm$ 0.91
FTAT w/o LCW	62.52 $\pm$ 0.86	61.73 $\pm$ 1.06	67.07 $\pm$ 1.20
FTAT	64.09 $\pm$ 1.14	63.64 $\pm$ 0.93	67.80 $\pm$ 2.71

learning rate without requiring tuning, indicating the advantage of our *Dynamic Model Ensembler* module.

Robustness of Batch Size. In the main experiments, the batch size of the data stream is set to 512 due to the large quantity of testing data. A natural question that arises is how the batch size affects the performance of the proposed method. We conduct experiments on the DIABETE dataset using an MLP backbone model, with batch sizes set to {64, 128, 256, 512, 1024}. As shown in Fig. 6, the results indicate that the accuracy and balanced accuracy metrics are robust across different batch sizes. Regarding the F1 score, it decreases as the batch size increases. Nevertheless, FTAT consistently outperforms existing methods by a margin.

Robustness of Hyperparameters. To validate whether our proposed FTAT approach is robust to the choices of hyperparameters, we conduct hyperparameter robustness experiments on DIABETE dataset evaluated by three metrics using MLP backbone model. Specifically, FTAT contains

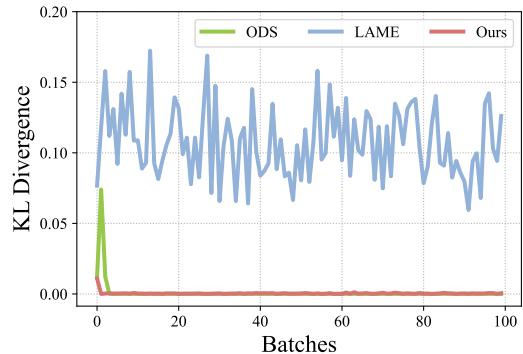


Figure 5: The performance of LAME, ODS, and FTAT in estimating label distribution evaluated using KL divergence.

three hyperparameters, i.e., ϵ , α and β . The hyperparameter α controls the rate at which the estimated label distribution is updated, enhancing the robustness of the FTAT to estimation errors in certain batches. The hyperparameter ϵ governs the entropy-based confident samples selection to accurately estimate the label distribution. β determines the construction of the neighbor set for entropy minimization, contributing to robust model adaptation. We conducted three runs of experiments for each set of hyperparameters with α in {0.08, 0.09, 0.10, 0.11, 0.15, 0.20}, $\epsilon = \text{Entropy}([p, 1 - p])$ where p was set to {0.72, 0.71, 0.70, 0.69, 0.65, 0.60}, and β in {0.28, 0.29, 0.30, 0.31, 0.40, 0.50}. Fig. 6 reports the average performance of each hyperparameter evaluated using three metrics on DIABETE dataset. The results demonstrate that FTAT is robust to slight changes in all hyperparameters.

5 Related Work

In this section, we mainly discuss two lines of related work, including test-time adaptation and deep tabular learning.

Test-time Adaptation. Test-time adaptation aims to adapt a source model to the distribution shift in testing data without using any source data. Previously, test-time training

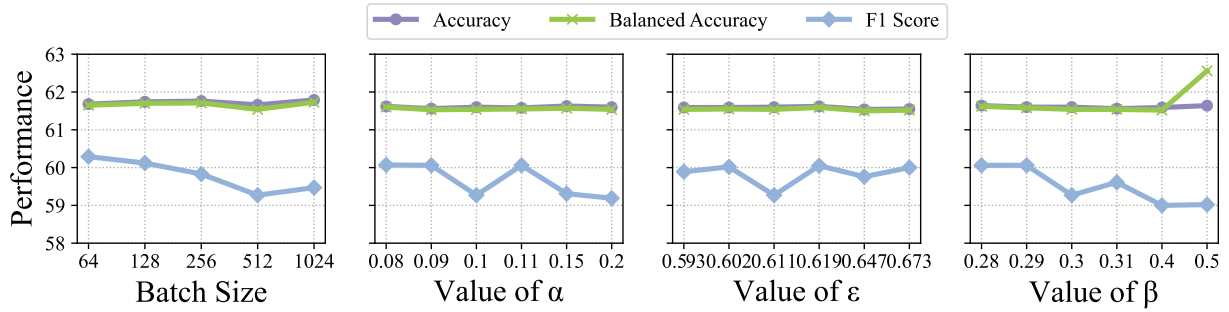


Figure 6: Robustness of batch size and hyperparameters α, ϵ, β on DIABETE dataset using MLP backbone model. The results indicate that minor perturbations to the hyperparameters of FTAT do not significantly affect its performance, demonstrating the practical robustness of FTAT.

Table 6: Performance of base models with different learning rates, directly ensemble of base models, and the FTAT approach on DIABETE dataset using the MLP backbone model. The best performance is in **bold**.

Method	Acc.	Balanced Acc.	F1
Lr=1e-3	61.49 $\pm$ 0.46	61.47 $\pm$ 0.42	60.39 $\pm$ 0.73
Lr=1e-4	61.58 $\pm$ 0.30	61.53 $\pm$ 0.26	59.37 $\pm$ 0.95
Lr=5e-4	61.56 $\pm$ 0.33	61.53 $\pm$ 0.30	59.93 $\pm$ 0.90
Lr=1e-5	61.58 $\pm$ 0.30	61.52 $\pm$ 0.26	59.23 $\pm$ 0.97
Average Ensemble	61.57 $\pm$ 0.31	61.51 $\pm$ 0.27	59.24 $\pm$ 0.95
FTAT	61.60 $\pm$ 0.31	61.53 $\pm$ 0.27	59.27 $\pm$ 0.96

studies, such as TTT (Sun et al. 2020) and TTT+ (Liu et al. 2021), manipulated the model in both the training and testing phases. They introduce self-supervised objectives at training time and adapt the model parameters by optimizing self-supervised objectives at testing time. However, when training data is inaccessible and model training cannot be controlled, test-time training paradigms become ineffective. Fully test-time adaptation aims to tackle this limitation by adapting the model without assumptions on the source model. Tent (Wang et al. 2021) updates the parameters of the BN layer at test time. EATA (Niu et al. 2022) additionally conducts active sample selection and weighting strategies for efficiency. Other studies (Gong et al. 2022; Goyal et al. 2022) also propose diverse methods to adapt the BN layer to the test data distribution to ensure performance. In practice, SAR (Niu et al. 2023) introduces a flat minimum optimization method to ensure generalization performance when the test batch size varies. CoTTA (Wang et al. 2022) works on continually non-i.i.d. scenarios using weight-averaged models, augmentation-averaged predictions, and stochastically restoring. LAME (Boudiaf et al. 2022) proposes a conservative approach to revise the model’s predictions instead of model parameters. ODS (Zhou et al. 2023) focuses on test-time adaptation settings where covariate and label distributions change together. Recent TTA studies mainly focus on images and natural language, paying little attention to tabular data. AdapTable (Kim et al. 2024) studies test-time adaptation for tabular data, effectively designing a graph-based module to address label shifts and providing insightful theoretical analyses. TabLog (Ren et al. 2024) is the first to

examine the structure of invariant rules for tabular data in the context of test-time adaptation. However, these studies require the training data to be available, which cannot be applied in our *AdaTab* problem setting and is not practical in real-world scenarios. Therefore, our paper focuses on the fully test-time adaptation problem for tabular data, an area that remains underexplored.

Deep Tabular Learning. Deep Tabular Learning aims to model tabular data for tasks such as classification and regression through deep learning methods. Unlike image and language data, the heterogeneity and high dimensionality make it difficult for models to extract spatial and semantic information. Recently, attention-based architectures have been introduced to the tabular data domain. FT-Transformer (Gorishniy et al. 2021) applies a feature tokenizer to heterogeneous feature columns and learns an optimal representation in embedding space. Additionally, TabTransformer (Huang et al. 2020), TabNet (Arik and Pfister 2021), and other deep tabular models (Badirli et al. 2020; Klambauer et al. 2017; Gorishniy, Rubachev, and Babenko 2022; Grinsztajn, Oyallon, and Varoquaux 2022) are proposed for better representation of tabular data. However, these methods typically works well in an i.i.d. setting, and may suffer from performance degradation when the test data distribution shifts.

6 Conclusion

In this paper, we investigate the problem of fully test-time adaptation for tabular data (*AdaTab*), an important and practically valuable issue that remains underexplored. Our observations highlight three key challenges in the *AdaTab* problem: the existence of label and covariate distribution shifts, the lack of effective data augmentation, and the sensitivity of model adaptation. To address these challenges, we propose the FTAT approach, which includes three novel modules: *Confident Distribution Optimizer*, *Local Consistent Weighter*, and *Dynamic Model Ensembler*. Our experimental results demonstrate that the FTAT approach outperforms existing FTTA methods, demonstrating its effectiveness in addressing tabular tasks.

One limitation of this paper is that the design of FTAT approach lacks deep theoretical understanding and we will explore in this direction in the future to provide deep insights for the following researchers.

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Fully Test-time Adaptation for Tabular Data

— Appendix —

The structure of Appendix is as follows:

- Section A provides detailed experimental setup and implementations.
- Section B presents detailed experimental results.
- Section C discusses the border impact.

A Detailed Experimental Setup

We have briefly introduced experimental setup in our main manuscript. Here, we provide detailed experimental setup and implementations in this section.

A.1 Details of Comparison Methods

To compare our FTAT approach with various FTTA algorithms, including typical FTTA methods, continual FTTA methods, and recently proposed robust FTTA methods. For typical TTA methods, we take two novel method into comparison:

- (1) TENT (Wang et al. 2021) updates the model parameters with entropy minimization loss;
- (2) EATA (Niu et al. 2022) performs activate sample selection for adaptation and Fisher regularization for anti-forgetting to achieve strong predicting performance.

For continual TTA methods, we compare with:

- (3) CoTTA (Wang et al. 2022) eliminates error accumulated in the data stream via weight-and-augmentation averaged pseudo-labels and parameters stochastic restoration. We adopt the perturb augmentation (Fang et al. 2022) as data augmentation methods for tabular data.

For robust TTA methods, we take various proposed methods into comparison:

- (4) SAR (Niu et al. 2023) conducts sample filtering based on test entropy and update model parameters to a flat minimum to achieve well and robust performance;
- (5) LAME (Boudiaf et al. 2022) modifies model output by adopting a conservative adaptation approach;
- (6) ODS (Zhou et al. 2023) decouples the mixed distribution shift and then addresses covariate and label distribution shifts accordingly.

A.2 Implementations Details

In this subsection, we provide the details of backbone model, configuration of training and testing phase to enhance the reproducibility. All experiments are conducted on a Linux server with one NVIDIA GeForce RTX 3050Ti GPU.

Backbone Models. For all experiments, we use three representative deep tabular models: MLP, Tabtransformer (Huang et al. 2020) and FT-Transformer (Gorishniy et al. 2021) as the backbone model.

Training Phase. For training the source model, we follow the TableShift benchmark (Gardner, Popovic, and Schmidt 2023) for all setting of training hyperparameters. Specifically, we train each backbone model with a batch size of 512 for several epochs, depending on the model’s convergence as evaluated on the validation set. The AdamW optimizer is used with a learning rate of 0.01 and a weight decay of 0.01.

Testing Phase. For test-time adaptation, we set the batch size to 512 due to the large number of samples in each test set. For the online model ensemble, we configure three base learners with different learning rates: $1e-5$, $5e-4$, and $1e-4$. We set the hyperparameters of FTAT to $\alpha = 0.1$, $\epsilon = 0.611$, $\beta = 0.3$ for all experiments to demonstrate its robustness to hyperparameters. The value of ϵ is computed by the function $\mathcal{E}(p) = \text{Entropy}([p, 1 - p])$, with p set to 0.7. For comparison methods, we use their original hyperparameters in their paper. We report mean $\pm$ stdev accuracy, balanced accuracy and F1 score over three runs using different random seeds.

Code and Datasets. The code and datasets used in this paper are available at <https://wnjxyk.github.io/FTTA>.

A.3 Dataset Details

We conduct experiments on tabular datasets with natural distribution shifts from the TableShift benchmark (Gardner, Popovic, and Schmidt 2023). We selected six datasets suffering notable performance degradation under distribution shifts and containing sufficient sample sizes for adaptation, to demonstrate the effectiveness of our FTAT approach. Overall, we conduct experiments with samples from 10K to 5M and features from 26 to 365, including HELOC, ANES, ASSISTMENTS, DIABETES, Hypertension and Health Ins, which are widely used in various applications. We give the details of each dataset as follows.

Table 7: Performance of FTAT approach and comparison methods on 6 datasets using TabTransformer backbone model. The best performance is in **bold**.

Method	HELOC			ANES			Health Ins.		
	Acc.	Balanced Acc.	F1	Acc.	Balanced Acc.	F1	Acc.	Balanced Acc.	F1
Non-Adaptation	55.66 ± 1.34	59.60 ± 1.11	44.30 ± 3.08	78.95 ± 0.27	75.30 ± 0.45	84.23 ± 0.12	65.35 ± 1.08	70.11 ± 0.67	65.88 ± 1.63
TENT	56.37 ± 1.55	59.60 ± 1.11	50.34 ± 5.91	78.82 ± 0.38	75.18 ± 0.56	84.14 ± 0.19	65.22 ± 1.18	70.07 ± 0.70	65.62 ± 1.82
EATA	56.37 ± 1.55	57.04 ± 4.04	52.57 ± 4.42	78.82 ± 0.38	75.18 ± 0.56	84.14 ± 0.19	65.35 ± 1.08	70.11 ± 0.67	65.88 ± 1.63
LAME	51.71 ± 4.68	59.56 ± 1.51	44.30 ± 6.64	78.57 ± 0.49	74.77 ± 0.80	84.04 ± 0.20	65.35 ± 1.08	70.11 ± 0.67	65.88 ± 1.63
CoTTA	56.37 ± 1.55	59.60 ± 1.11	50.34 ± 5.91	78.82 ± 0.38	75.18 ± 0.56	84.14 ± 0.19	65.35 ± 1.08	70.11 ± 0.67	65.88 ± 1.63
ODS	52.19 ± 4.15	56.77 ± 3.56	44.70 ± 6.56	78.48 ± 0.70	74.69 ± 1.09	83.97 ± 0.31	57.14 ± 5.22	64.75 ± 3.23	51.33 ± 10.03
SAR	56.37 ± 1.55	59.60 ± 1.11	50.34 ± 5.91	78.82 ± 0.38	75.18 ± 0.56	84.14 ± 0.19	65.35 ± 1.08	70.11 ± 0.67	65.88 ± 1.63
FTAT	65.01 ± 1.09	64.34 ± 1.05	69.06 ± 1.11	80.00 ± 0.19	79.07 ± 0.14	83.32 ± 0.25	69.95 ± 0.44	61.27 ± 0.93	79.77 ± 0.10

Method	ASSIST			DIABETE			Hypertension		
	Acc.	Balanced Acc.	F1	Acc.	Balanced Acc.	F1	Acc.	Balanced Acc.	F1
Non-Adaptation	49.04 ± 4.23	53.07 ± 2.12	58.92 ± 2.34	60.85 ± 0.32	60.59 ± 0.34	50.00 ± 1.44	54.87 ± 0.89	59.24 ± 0.63	46.25 ± 2.16
TENT	47.83 ± 3.39	53.11 ± 2.18	61.27 ± 0.32	60.85 ± 0.06	60.59 ± 0.06	50.00 ± 0.11	41.74 ± 0.01	50.12 ± 0.01	0.78 ± 0.06
EATA	45.28 ± 0.16	51.44 ± 0.14	61.40 ± 0.41	60.85 ± 0.06	60.59 ± 0.05	50.00 ± 0.09	54.86 ± 0.89	59.24 ± 0.63	46.24 ± 2.15
LAME	45.12 ± 3.46	51.30 ± 2.20	61.40 ± 0.41	61.19 ± 0.04	60.96 ± 0.05	51.76 ± 0.31	54.87 ± 0.89	59.24 ± 0.63	46.25 ± 2.16
CoTTA	45.51 ± 0.39	51.64 ± 0.34	61.56 ± 0.16	60.85 ± 0.06	60.59 ± 0.06	50.00 ± 0.11	54.87 ± 0.89	59.24 ± 0.63	46.25 ± 2.16
ODS	45.12 ± 3.46	51.30 ± 2.20	61.40 ± 0.41	61.23 ± 0.11	60.99 ± 0.12	51.83 ± 0.43	54.87 ± 0.89	59.24 ± 0.63	46.25 ± 2.16
SAR	45.12 ± 3.46	51.30 ± 2.20	61.40 ± 0.41	60.85 ± 0.07	60.59 ± 0.06	50.00 ± 0.11	54.87 ± 0.89	59.24 ± 0.63	46.25 ± 2.16
FTAT	54.81 ± 1.80	55.33 ± 0.59	53.20 ± 2.62	61.74 ± 0.27	61.64 ± 0.27	57.95 ± 0.38	62.88 ± 0.25	60.72 ± 0.27	69.82 ± 0.67

Table 8: Performance of FTAT approach and comparison methods on 6 datasets using FT-Transformer backbone model. The best performance is in **bold**.

Method	HELOC			ANES			Health Ins.		
	Acc.	Balanced Acc.	F1	Acc.	Balanced Acc.	F1	Acc.	Balanced Acc.	F1
Non-Adaptation	46.26 ± 1.05	52.48 ± 0.78	13.32 ± 4.50	75.47 ± 1.31	71.50 ± 2.06	81.80 ± 0.45	58.33 ± 4.05	65.44 ± 2.56	54.06 ± 7.72
TENT	44.98 ± 1.83	51.45 ± 1.39	8.11 ± 7.80	63.02 ± 0.89	54.52 ± 1.15	76.19 ± 0.40	36.44 ± 0.03	50.05 ± 0.02	0.24 ± 0.11
EATA	45.95 ± 1.00	52.23 ± 0.74	12.27 ± 4.38	74.65 ± 1.66	70.16 ± 2.49	81.51 ± 0.67	57.40 ± 4.35	64.86 ± 2.79	52.23 ± 8.49
LAME	43.14 ± 0.04	50.03 ± 0.03	0.20 ± 0.14	75.37 ± 1.34	71.35 ± 2.12	81.73 ± 0.48	59.08 ± 4.57	65.44 ± 2.60	55.91 ± 9.04
CoTTA	46.26 ± 1.05	52.48 ± 0.78	10.67 ± 7.65	75.47 ± 1.31	71.50 ± 2.06	81.80 ± 0.45	58.33 ± 4.05	65.44 ± 2.56	54.06 ± 7.72
ODS	43.14 ± 0.04	50.03 ± 0.03	0.20 ± 0.14	75.41 ± 1.39	71.41 ± 2.15	81.75 ± 0.54	59.99 ± 3.45	65.54 ± 1.53	58.37 ± 6.92
SAR	43.30 ± 0.00	50.20 ± 0.00	30.60 ± 0.00	75.47 ± 1.31	71.50 ± 2.06	81.80 ± 0.45	58.33 ± 4.05	65.44 ± 2.56	54.06 ± 7.72
FTAT	65.09 ± 0.27	63.76 ± 0.39	70.49 ± 1.25	72.58 ± 0.18	74.40 ± 0.28	73.73 ± 0.21	63.78 ± 0.07	50.26 ± 0.11	77.82 ± 0.03

Method	ASSIST			DIABETE			Hypertension		
	Acc.	Balanced Acc.	F1	Acc.	Balanced Acc.	F1	Acc.	Balanced Acc.	F1
Non-Adaptation	58.32 ± 0.07	62.99 ± 0.09	67.63 ± 0.09	60.86 ± 0.17	60.67 ± 0.19	53.21 ± 1.34	58.88 ± 0.28	61.84 ± 0.14	55.71 ± 0.79
TENT	58.25 ± 0.07	62.91 ± 0.09	67.57 ± 0.09	59.97 ± 0.33	59.68 ± 0.35	47.12 ± 1.48	47.01 ± 6.39	53.88 ± 4.52	18.83 ± 20.89
EATA	48.04 ± 0.25	53.85 ± 0.19	62.60 ± 0.07	51.33 ± 0.47	50.74 ± 0.46	3.47 ± 2.12	58.84 ± 0.27	61.82 ± 0.13	55.62 ± 0.78
LAME	56.54 ± 2.48	62.98 ± 0.06	67.63 ± 0.07	60.52 ± 0.19	60.28 ± 0.20	50.23 ± 0.91	58.78 ± 0.27	61.78 ± 0.15	55.47 ± 0.74
CoTTA	58.25 ± 0.07	62.91 ± 0.09	67.57 ± 0.09	60.64 ± 0.14	60.40 ± 0.15	50.67 ± 0.71	58.88 ± 0.28	61.84 ± 0.14	55.71 ± 0.79
ODS	57.39 ± 1.19	62.16 ± 1.04	67.14 ± 0.58	59.60 ± 0.52	59.29 ± 0.55	45.53 ± 2.23	58.77 ± 0.26	61.78 ± 0.14	55.45 ± 0.72
SAR	58.25 ± 0.07	62.91 ± 0.09	67.57 ± 0.09	60.63 ± 0.14	60.39 ± 0.15	50.67 ± 0.71	59.64 ± 0.65	62.24 ± 0.29	57.52 ± 1.63
FTAT	58.62 ± 0.24	63.13 ± 0.22	67.55 ± 0.14	61.16 ± 0.09	61.17 ± 0.09	61.25 ± 0.31	62.81 ± 1.22	62.50 ± 0.57	66.52 ± 3.98

HELOC. The Home Equity Line of Credit (HELOC) is a part of credit secured by the applicant’s home. A HELOC provides access to a revolving credit line with a lower interest rate than other types of loans. To assess an applicant’s suitability for a HELOC, a lender evaluates an applicants’ financial background to predict whether a given applicant is likely to repay a line of credit and, if so, how much credit should be extended. In order to accurate credit risk predictions for their overall utility for both lenders and borrowers and achieve equal treatment, HELOC dataset which contains 10,459 samples need to predict whether a consumer is 90 days past due or worse at least once over a period of 24 months using 38 features varying from financial activity to credit inquiries.

ANES. Understanding participation in elections is a critical task for policymakers, politicians, and those with an interest in democracy. Predicting which individuals will vote in an electio, is widely acknowledged as critical to polling and campaigning in U.S. politics. For better understanding individual activity in presidential election, ANES dataset predict whether an individual will vote in the U.S presidential election using 365 features including voting behavior, elections, public opinion and attitudes. The ANES dataset includes 60,377 samples.

ASSIST. ASSISTMENTS dataset in education field (abbreviated as ASSIST). The ASSISTments tutoring platform is a free, web-based, data-driven tutoring platform for students in grades 3-12. ASSISTMENTS dataset contains affect predictions such as such as boredom, confusion, frustration, and engaged problem-solving behavior on students who use the ASSISTMENTS tutoring platform. The numbers of features and samples in ASSISTMENTS dataset are 26 and 2,667,776.

DIABETES. Effective management and treatment of diabetic patients admitted to the hospital can have a significant impact on their health outcomes (Umpierrez et al. 2002). Several factors can affect the quality of treatment patients receive (Strack et al. 2014). One of the costliest and potentially most adverse outcomes after a patient is released from the hospital is for that patient to be readmitted soon after their initial release. Thus, predicting the readmission of patients is a priority from both a medical and economic perspective. DIABETE dataset which contains 1,444,176 samples predict whether a diabetic patient is readmitted to the hospital within 30 days of their initial release using 183 related features.

Hypertension. Hypertension, or systolic blood pressure (typically systolic pressure 130 mm Hg or higher or diastolic 80 or higher) affects nearly half of Americans. When left untreated, hypertension is associated with the strongest evidence for causation of all risk factors for heart attack and other cardiovascular disease (Fuchs and Whelton 2020). As a result, it is important to predict blood pressure accurately and efficiently. Hypertension dataset has a goal to achieve effective blood pressure measurement and increase the prediction accuracy. Hypertension dataset contains 846,781 samples with 100 features related to several risk factors for hypertension.

Health Ins. Public health insurance makes a significant performance in providing affordable and accessible medical care for individuals. A high level of health insurance ownership is important for the healthy development of the individual. So, it is important to raise the rate of owning health insurance. Health Ins. dataset is related to public coverage field and the goal is to predict whether an individual is covered by public health insurance using 135 features. The number of samples in Health Ins. dataset is 5,916,565.

B Detailed Experimental Results

We neglect the detailed experimental results of TabTransformer (Huang et al. 2020) and FT-Transformer (Gorishniy et al. 2021) backbone models in our main manuscript due to the space limit. The neglected results are presented in Tab. 8 and Tab. 7. The results show that our FTAT approach gives the best performance on majority of datasets, demonstrating the effectiveness of our approach. In the remaining cases, FTAT approach achieves competitive performance compared to the best-performing method.

C Broader Impact

This paper aims to advance the field of deep tabular learning by addressing the negative effects of distribution shifts that occur during the testing phase. We believe our work has many potential societal impacts, the majority of which are positive and none of which need to be highlighted here.